

Empowering the Public: The Growing Role of Citizen Science

Open Asia October 2026

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Co-funded by the European Union

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WHO ARE WE?



Prof.dr. Marjolein Zweekhorst

Dr. Sarju Sing Rai



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VU University, Amsterdam, The Netherlands

- 32.000 students
- 45 BSc programs
- 100 MSc programs
- 4700 staff
- 31 % female professors
- 358 phd defences / year







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VU University

A Broader Mind



At VU Amsterdam you don't just become something, you become someone. Develop 'A Broader Mind!'



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OPEN SCIENCE

Source: Unesco

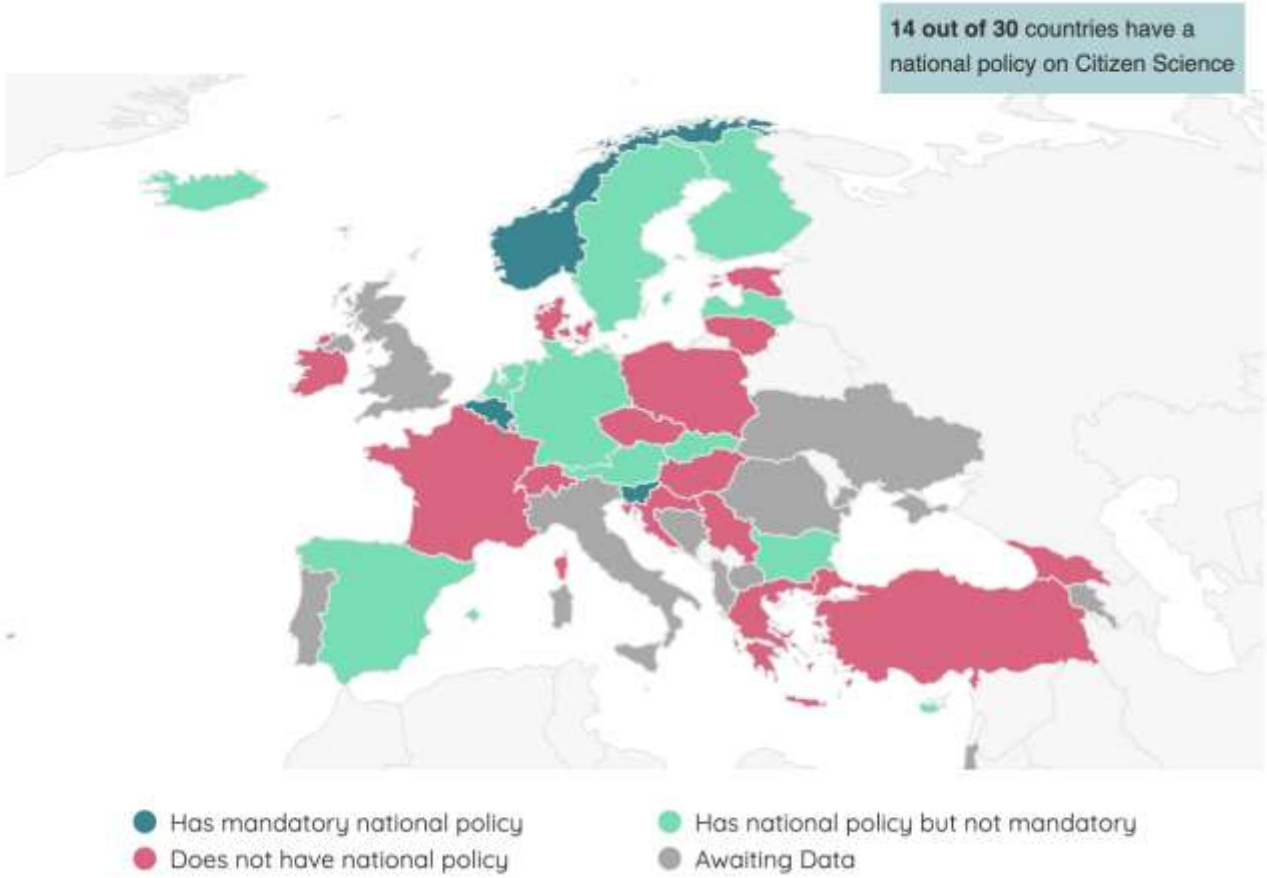
Understanding open Science 2022

https://doi.org/10.54677/UTCD9302





National Policy on Citizen Science



Data source: Survey on National Contributions to EOSC and Open Science 2024.

General Overview of Citizen Science in Europe for 2024

New Citizen Science project in EU Countries

0 projects

↓ -100% from last year



EU Countries with Policy on Citizen Science

47%

↑ +5% from last year



Financial investments in Citizen Science

16.91 mil Euro

↑ +8% from last year



EU Countries with Financial Strategy on Citizen Science

13%

→ 0% from last year



EU Countries with Citizen Science Monitoring Initiatives

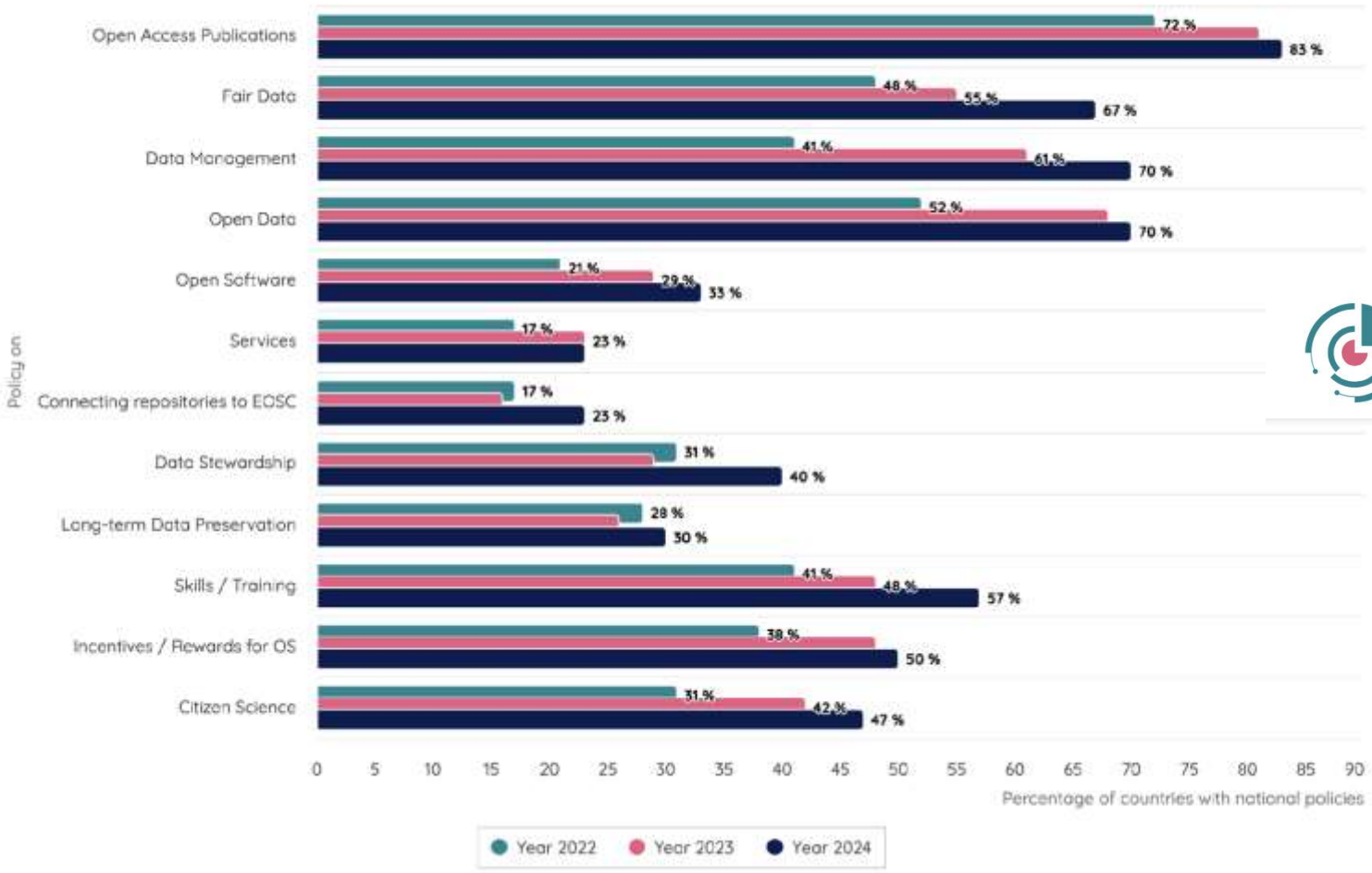
10%

↑ +4% from last year



Data source: Survey on National Contributions to EOSC and Open Science 2024 - showing data for 2024, except financial contributions, which reflect the previous year (2023).

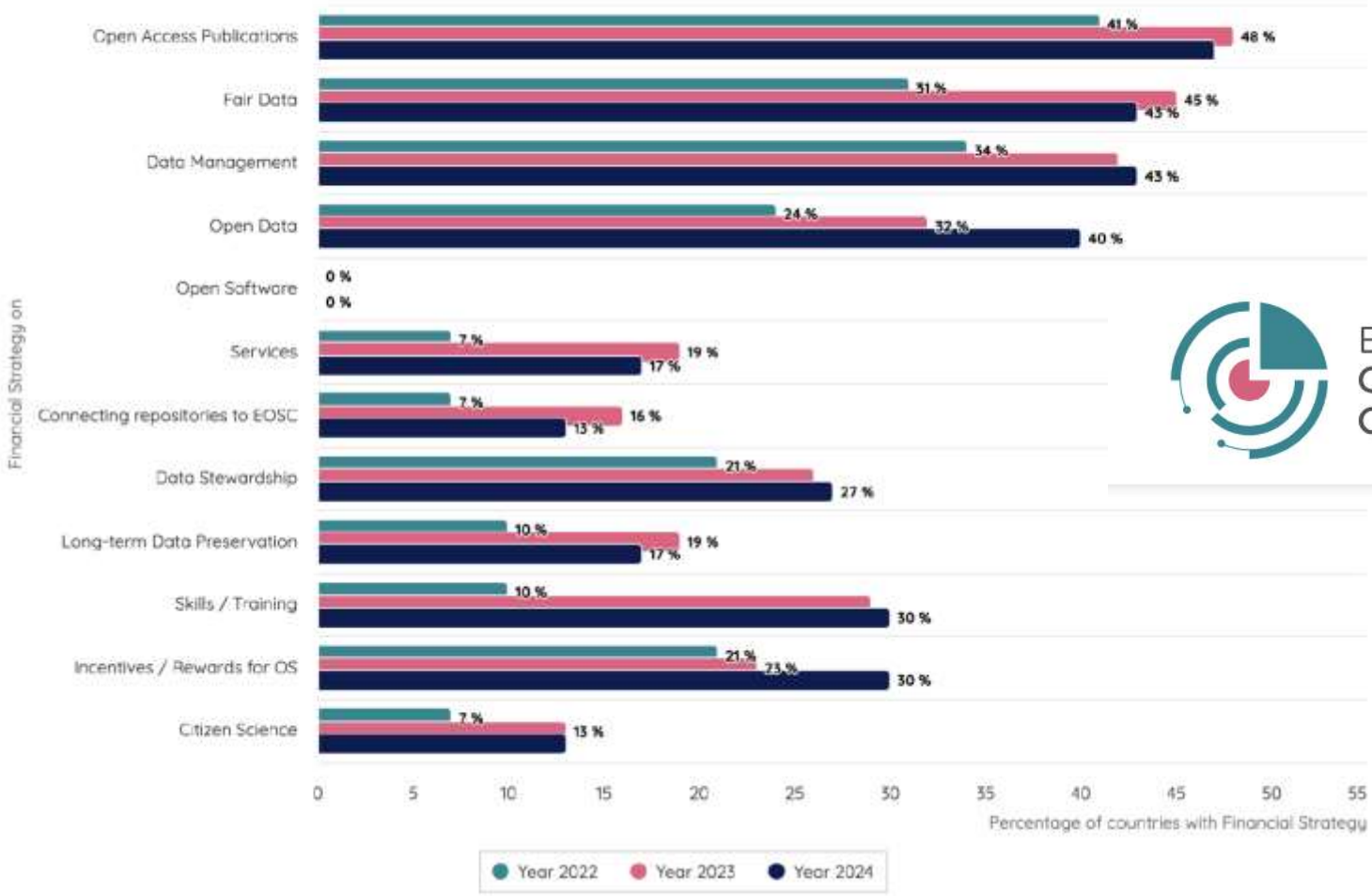
Trends in National Open Science Policies by Areas



EOSC
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Trends in Financial Strategy on EOSC and Open Science by Areas



EOSC
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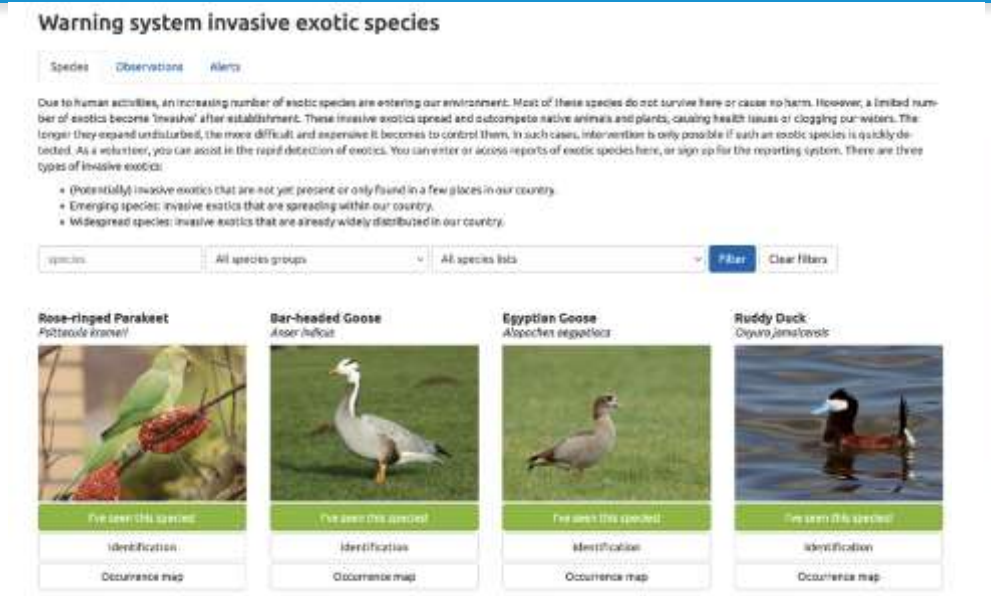
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Open Science: Citizen Science different models / definitions

Source	Definition	Main idea
Bonney et al (2009)	Citizens as data collectors	Data collection
Irwin (1995)	Science for and by citizens	Accountability and participation
European Citizen Science Association 2015	Public and researchers	Joined collaboration
Haklay (2013)	Different levels of participation	crowdsourcing, distributed intelligence, participatory science, and extreme citizen science
UNESCO (2021)	inclusive	Participtation throughout the research cycle

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Citizen science contributory model





• Citizens mainly contribute observations or measurements, while professionals design the research and analyze the data.

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Citizen science colloborative model



ReefCheck Malaysia

- Citizens mainly contribute observations or measurements, while professionals design the research and analyze the data.





- Citizens contribute data but are also involved in
- shaping research questions or methods.



Samen Meten Dataportaal Nederland

Dustboy Thailand

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Citizen science: co-created / participatory model







Citizens and researchers jointly design, implement, and sometimes even publish research.
Focus is often on **local issues, empowerment, and policy influence**.





- Stigma-reduction interventions in Indonesia
- Mental health interventions in India, for migrants in Europe
- Maternal health in LMICs – Safe Motherhood, teenage pregnancies, anti-natal care
- Sexual and reproductive health in Laos – health literacy







Citizen science: grasssroots



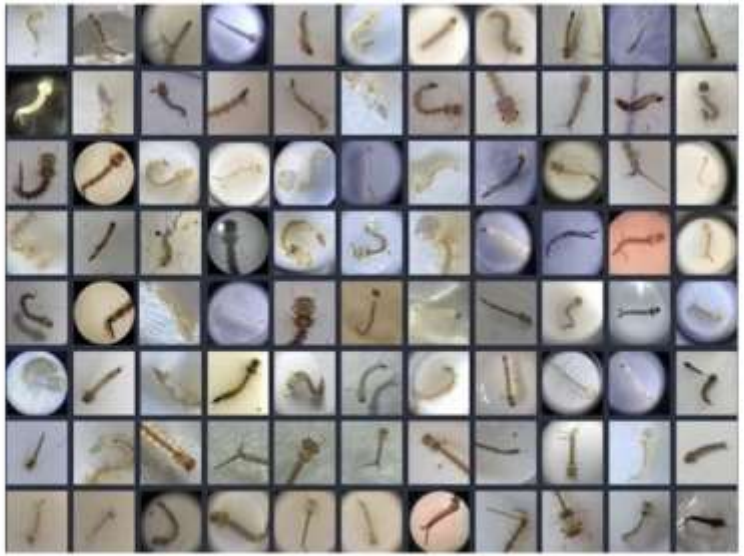
Home Meten en observeren Resultaten Blog FAQ Achtergrond Steden Agenda Doe mee!

- Citizen-driven, often outside institutional science, with goals ranging from **local activism** to **open innovation**.

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Citizen science: international projects





This image, selected as Biomedical Picture of the Day, shows mosquito larvae photos submitted by citizen scientists during the Mosquito Habitat Photo Challenge, July-August, 2021. Collage created by Kristen Weaver.

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Citizen science in india and Malaysia







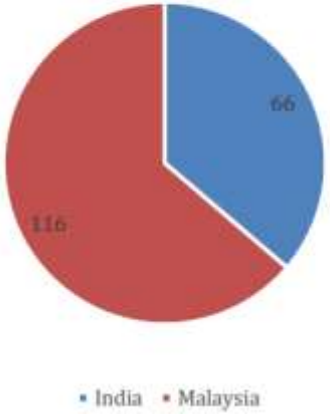
OPEN-ASIA: CS Survey

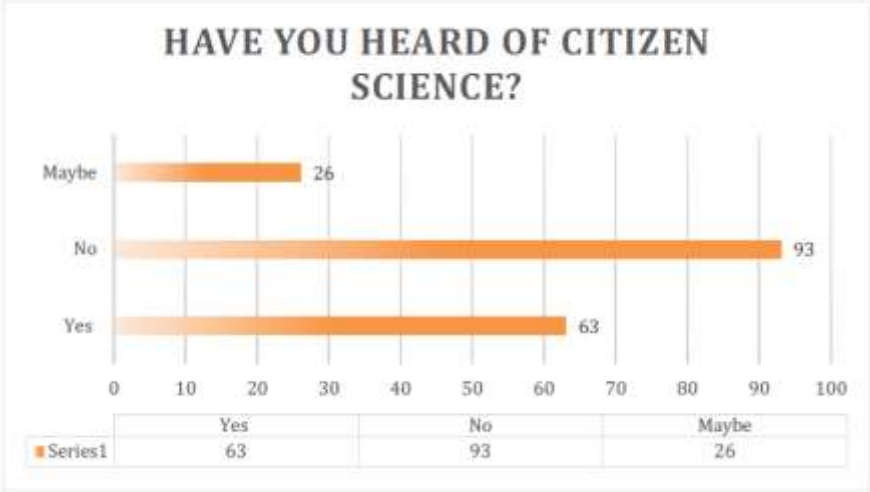
Co-funded by the Erasmus+ Programme of the European Union

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Preliminary Insights from the Public Survey: Citizen Science and Public Engagement Practices in Higher Education

Countries reprsented by the survey participants



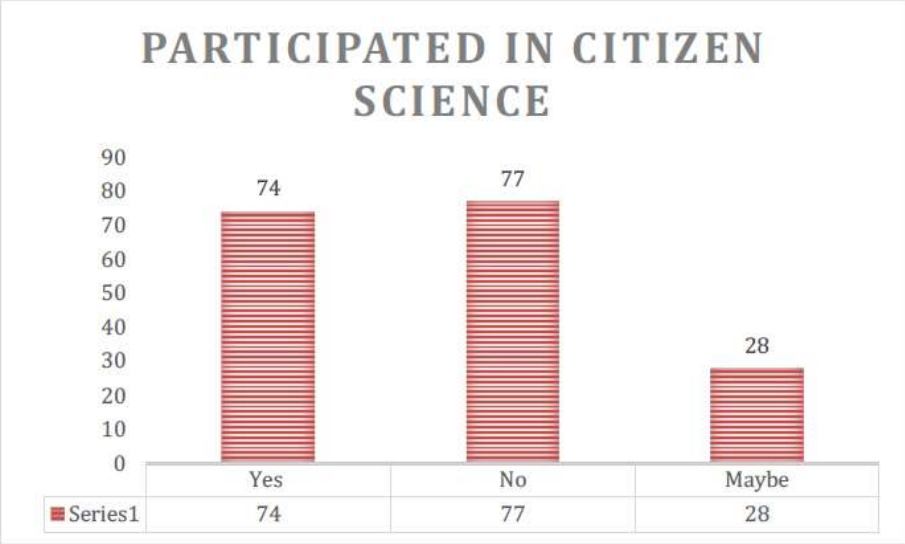


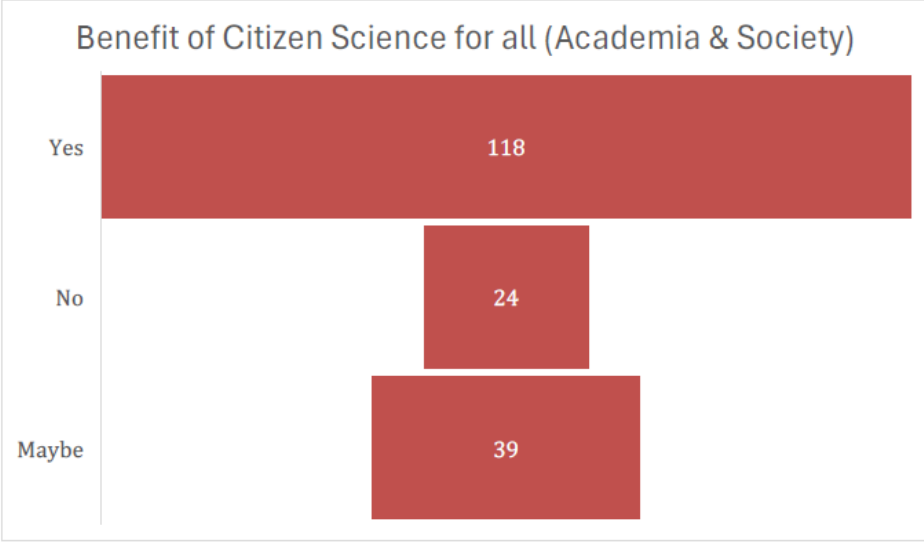
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OPEN-ASIA: CS Survey

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Citizen Science (CS) Policy in the Netherlands

- *"Slow and Steady"* process
- National Plan for Open Science in 2017
- National Program for Open Science (NPOS) – 2019
- **A bottom-up initiative**: Citizen Science practitioners formed a Working group in 2020 to embed CS in NPOS
- 2021 – CS Integrated in NPOS
- 2022 – First national network for CS



National Programme for Open Science – Ambition 2030

- In response to the UNESCO Recommendation on Open Science (2021) – NPOS 2030
- GOAL: "Close collaboration between knowledge institutions, government, industry, and citizens to strengthen the international position of Dutch science and optimize the processes of creating, sharing, and communicating knowledge for the benefit of society.“
- Action Line: "**Towards Societal Engagement and Citizen Science**".



NPOS2030 – Citizen Science Policy

- **Raise awareness and understanding** of CS value across society
- Consolidate and develop **best practices**, encouraging high quality, impactful projects through infrastructure, guidelines and collaborations
- **Build capacity** through training, educational resources, and institutional support
- **Strengthen infrastructure** – funding, mechanisms, technical tools, networks, and platforms to support CS
- Embed CS in **recognition and reward systems**, ensuring participatory research is valued



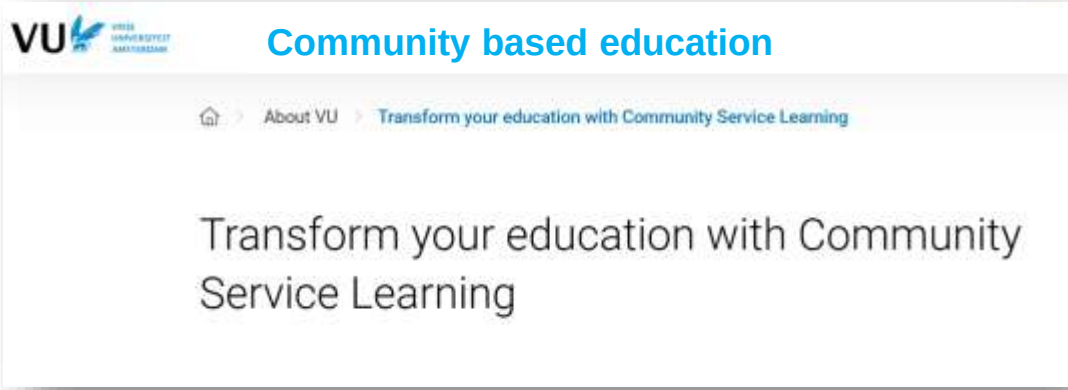
Citizen Science Practices Acculturation

Community Based Participatory research



THE SCIENCE SHOP

Community based education



Public engagement



LIVING LAB

Community engagement in Science/Research



Citizen Science Practices & Acculturation: Drivers

Blog: over 20 million euro awarded in 2024

14 January 2025

2024 was a landmark year for Open Science NL, with over 20 million euros awarded to advance open science initiatives across the Netherlands. From strengthening digital competence centres to fostering citizen science and open research infrastructure, the first-ever Open Science NL Work Programme made significant strides. As we enter 2025, let's look back at the impact we made and look forward to what's next.



Citizen Science NOW

NETWORK • TRANSFORM • GROW

OPEN SCIENCE NL

Citizen Science Hubs

ecsa 2026 conference

Learnings from our Experience(s)

- **Bottom-up initiatives & Top-down commitments**
- Raising awareness; Highlighting excellent practices/examples
- Defining goals, standards, guidelines
- Endorsing and Embedding OS/CS in science system(s)
- National and regional networks
- Integration into existing priorities/developments



THANK YOU FOR YOUR ATTENTION]



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Open Science: Citizen Science different models / definitions

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